

Lab_Syn_3.1

You have to synthesize **ALU_Top hierarchy** files using **Design Compiler shell** and generate **Technology Dependent** gate level netlists in Verilog format using standard cell libraries (typical, worst, best) inside **std_cells** folder:-

- **scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.db**
- **scmetro_tsmc_cl013g_rvt_ff_1p32v_m40c**
- **scmetro_tsmc_cl013g_rvt_ss_1p08v_125c**

Steps: -

1. Copy **Lab_Syn_3.1** folder from your computer into the virtual machine inside **Labs** directory
2. Open **syn_script.tcl** and **Update variable "flattened" to 0** and run **syn_script.tcl** script through design compiler shell using the following command in the terminal `>> dc_shell -f syn_script.tcl | tee syn.log`
 - Open the GUI and view the schematic
3. Open **syn_script.tcl** and **Update variable "flattened" to 1** and run **syn_script.tcl** script through design compiler shell using the following command in the terminal `>> dc_shell -f syn_script.tcl | tee syn.log`
 - Open the GUI and view the schematic
4. What is the difference between the **2 netlists** **ALU_mapped_flattened.v** & **ALU_mapped_hierarchy.v** and which command make this difference ?
5. View the **2 Area reports** **area_flattened.v** Verses **area_hierarchy.v**